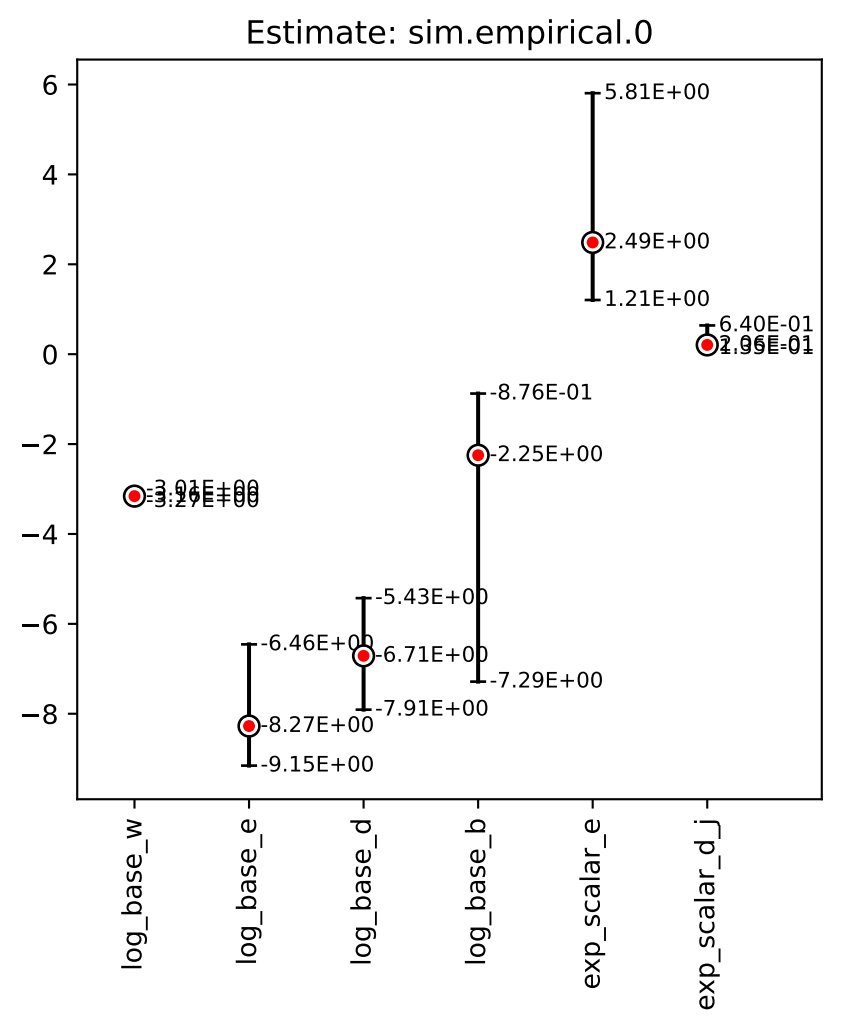
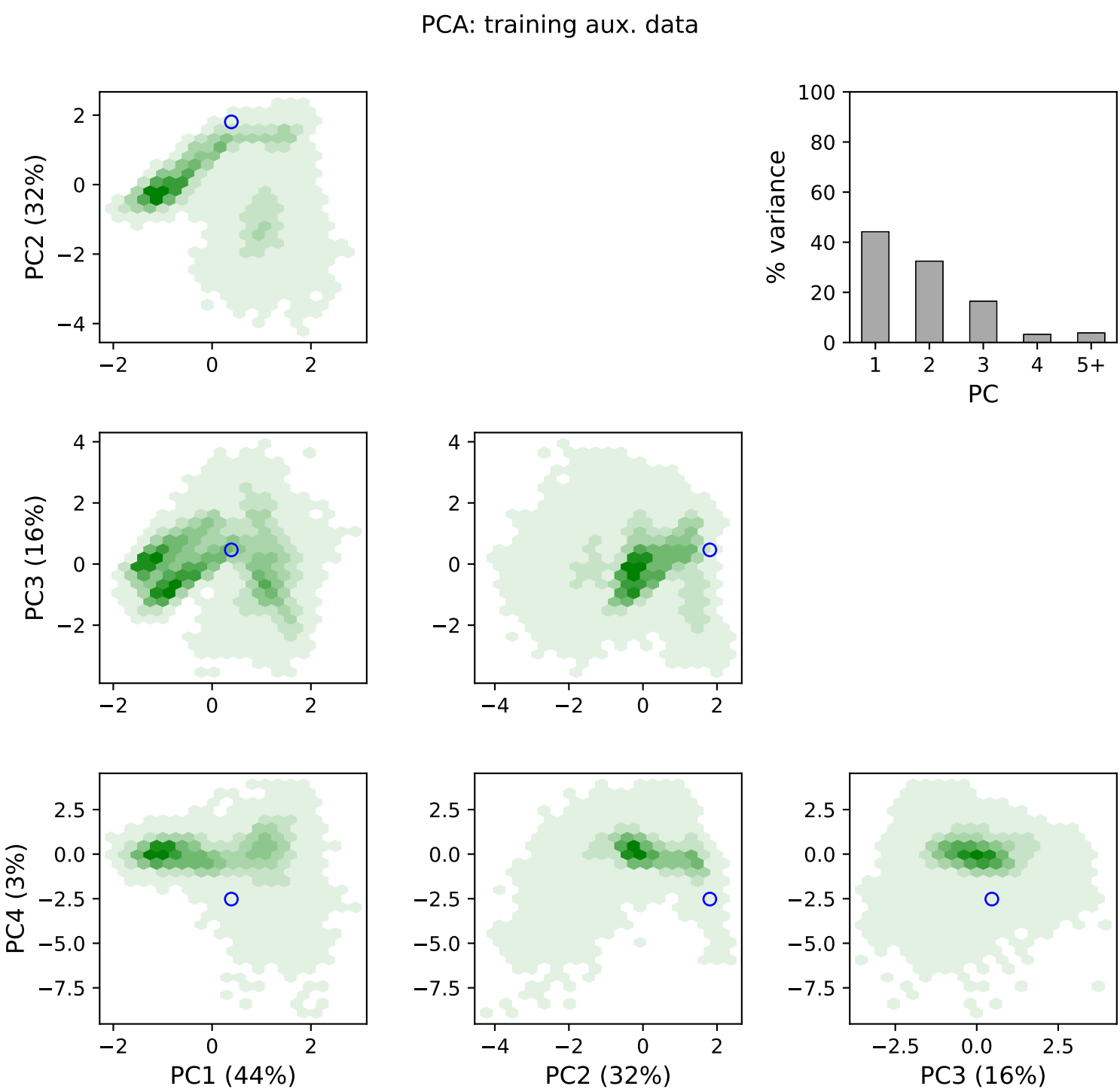
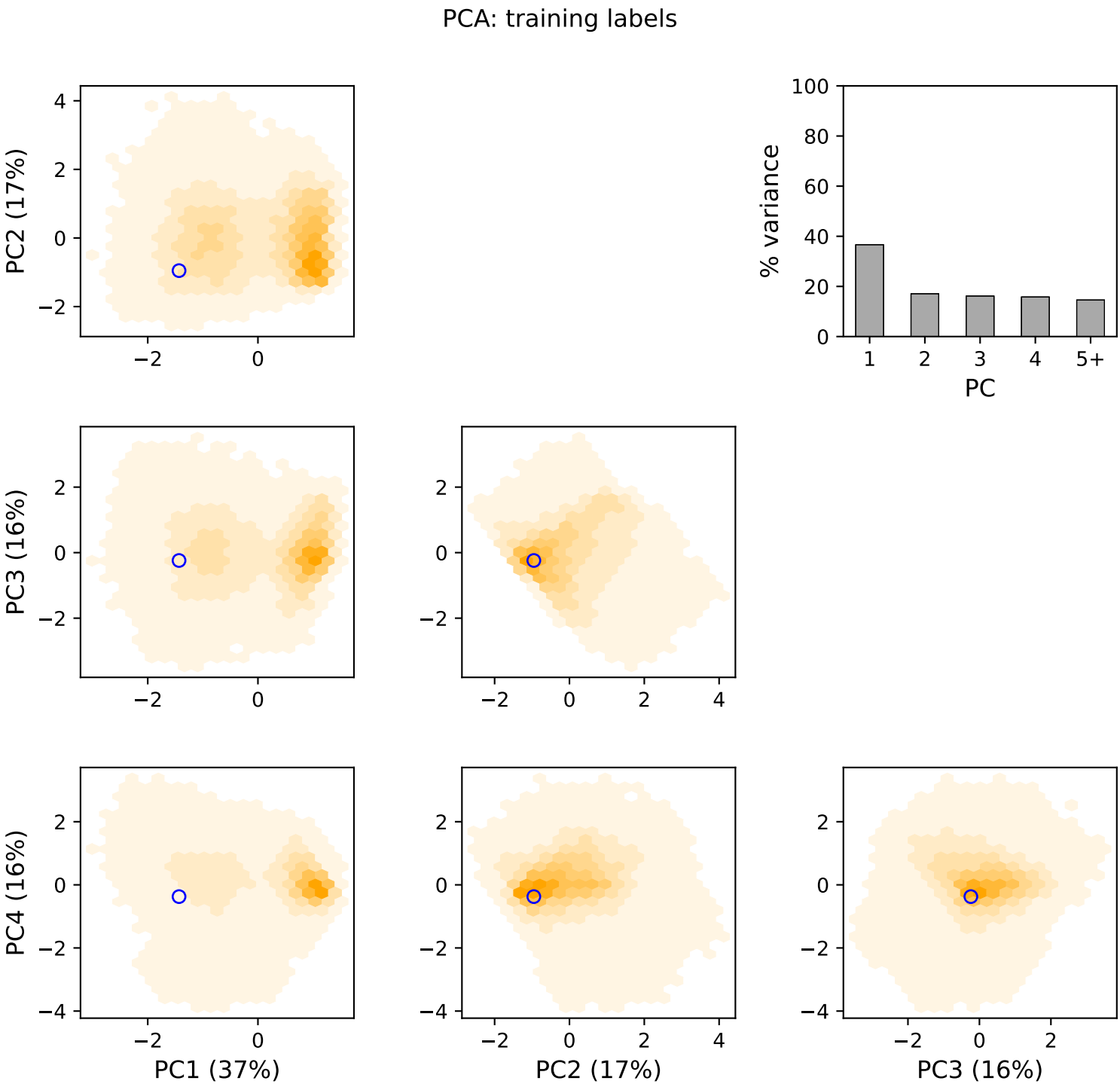
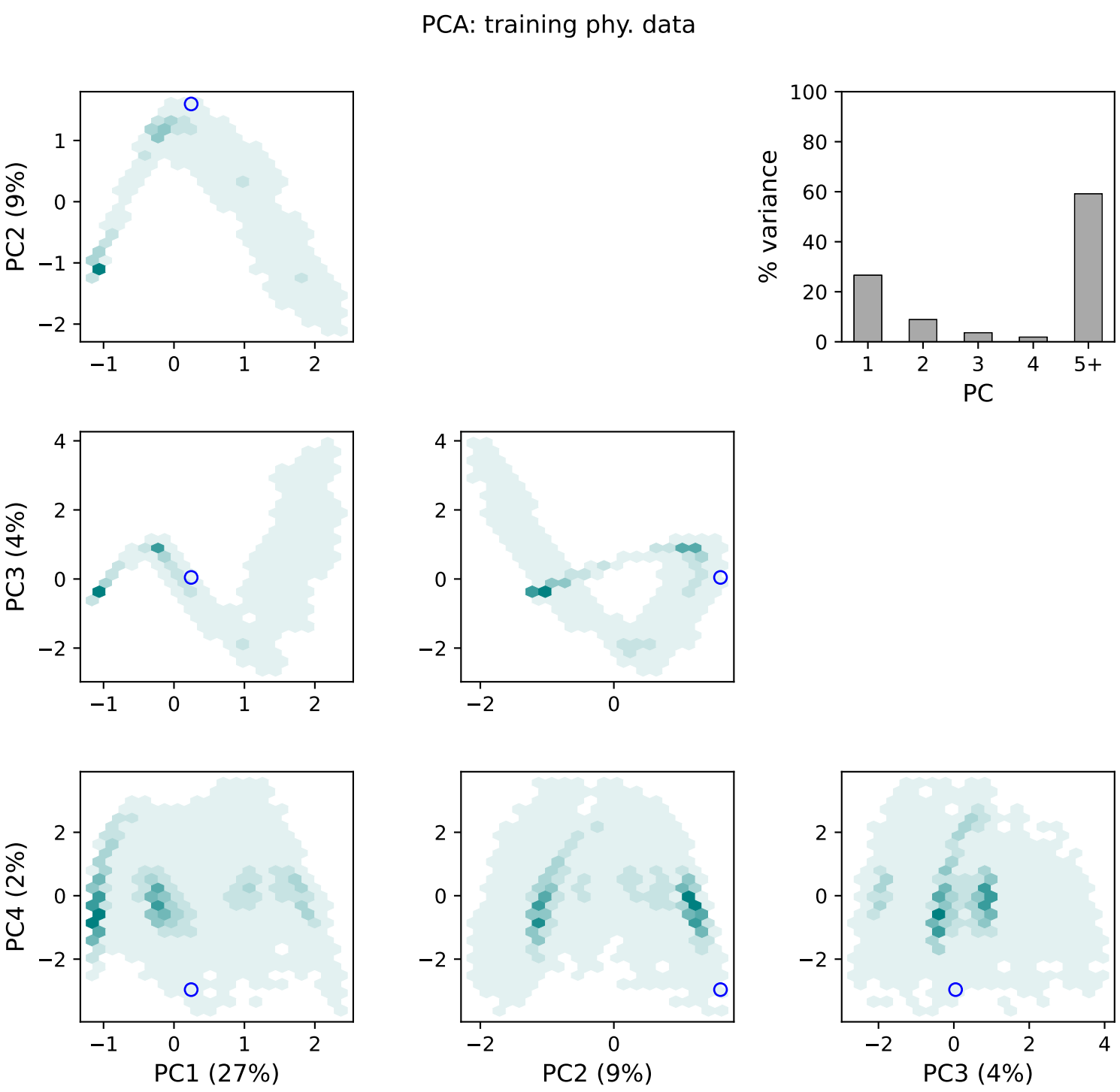
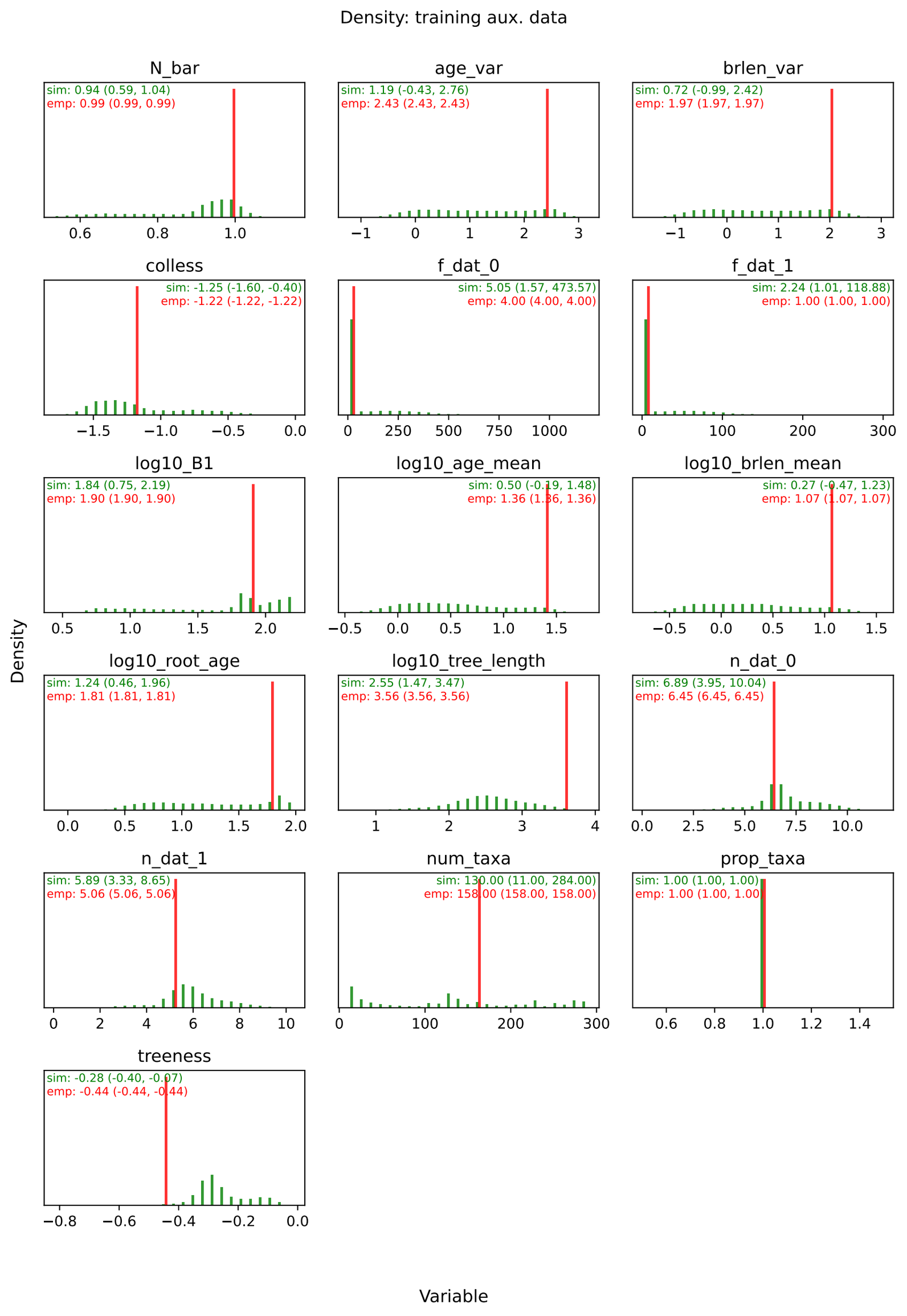
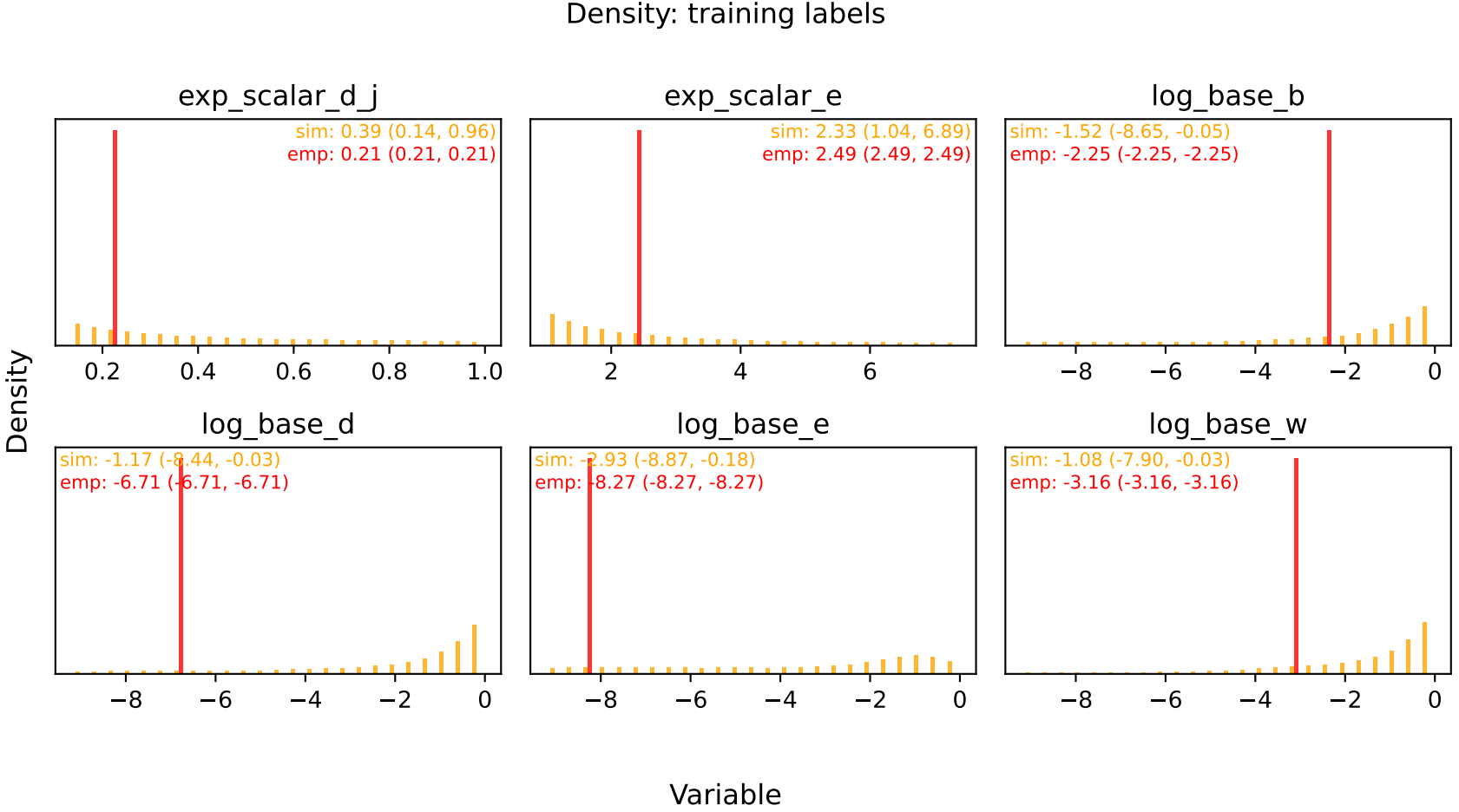

Estimate: sim.empirical.0
6
4
2
0
−2
−4
−6
−8
5.81E+00
2.49E+00
1.21E+00
6.40E-01
-8.76E-01
-2.25E+00
-5.43E+00
-6.46E+00
-6.71E+00
-7.29E+00
-7.91E+00
-8.27E+00
-9.15E+00
log_base_w
log_base_e
log_base_d
log_base_b
exp_scalar_e
exp_scalar_d_j

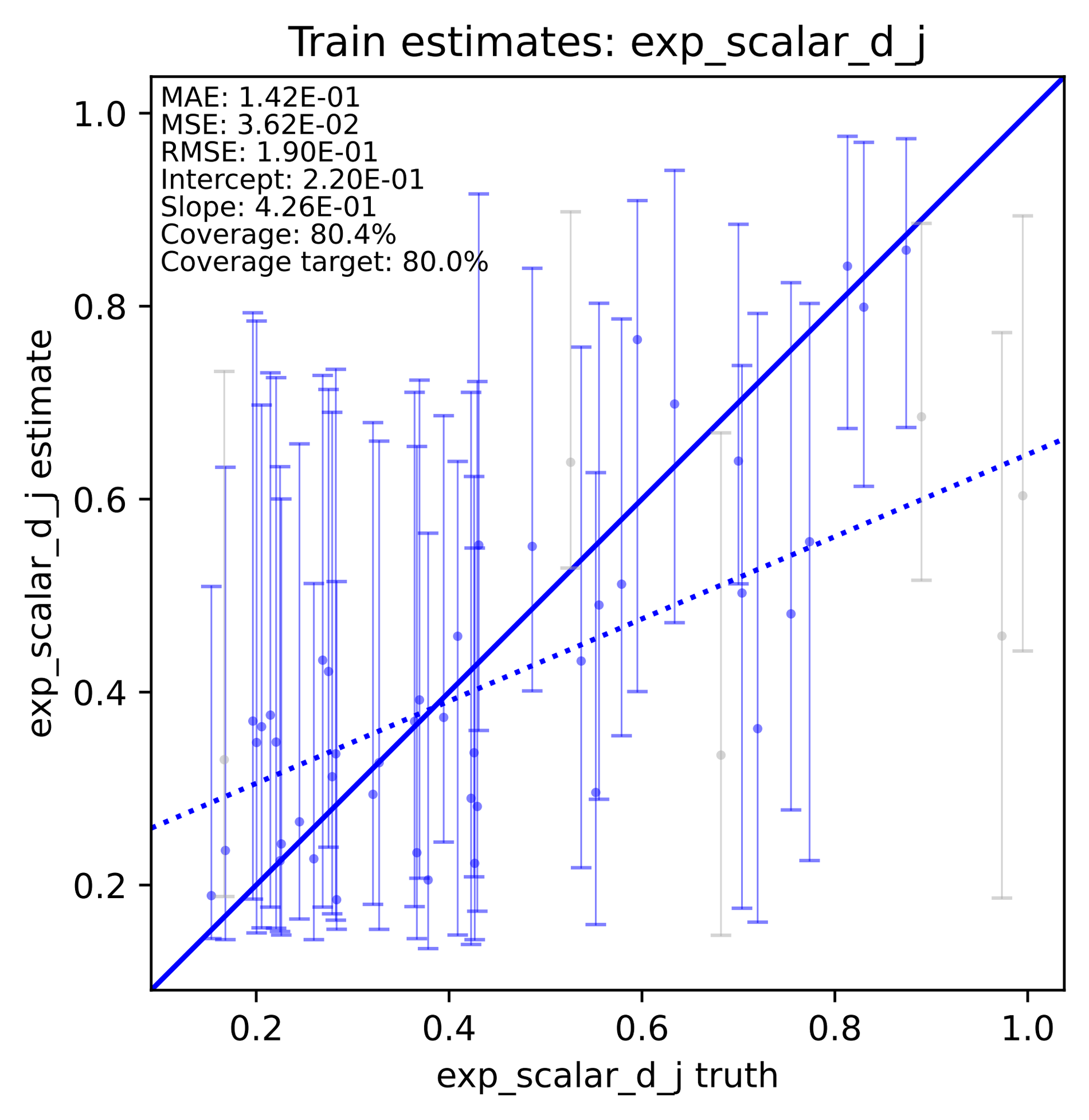

PCA: training aux. data
PC2 (32%)
% variance
PC
PC3 (16%)
PC4 (3%)
PC1 (44%)
PC2 (32%)
PC3 (16%)

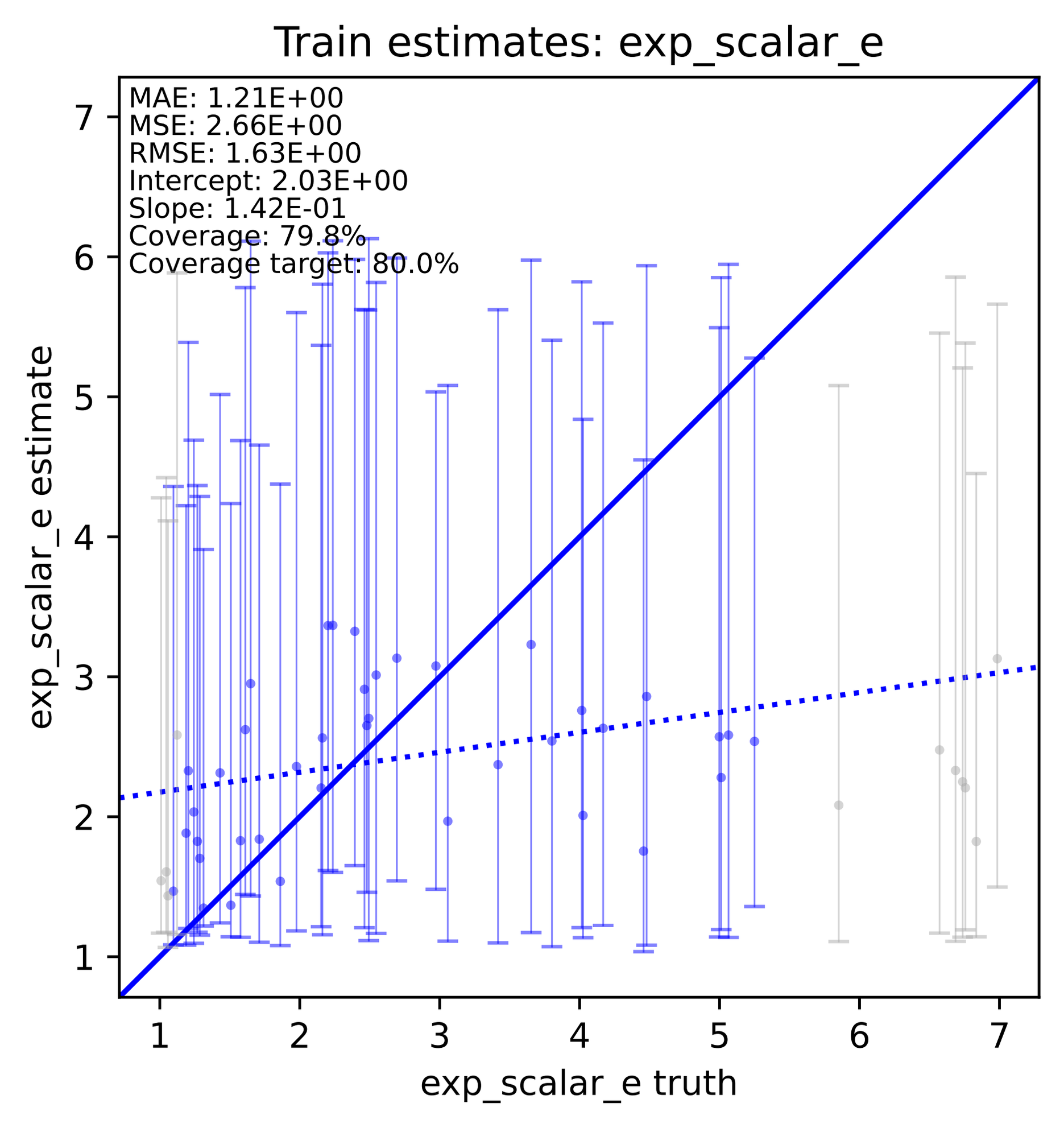

PCA: training labels
PC2 (17%)
% variance
PC
PC3 (16%)
PC4 (16%)
PC1 (37%)
PC2 (17%)
PC3 (16%)

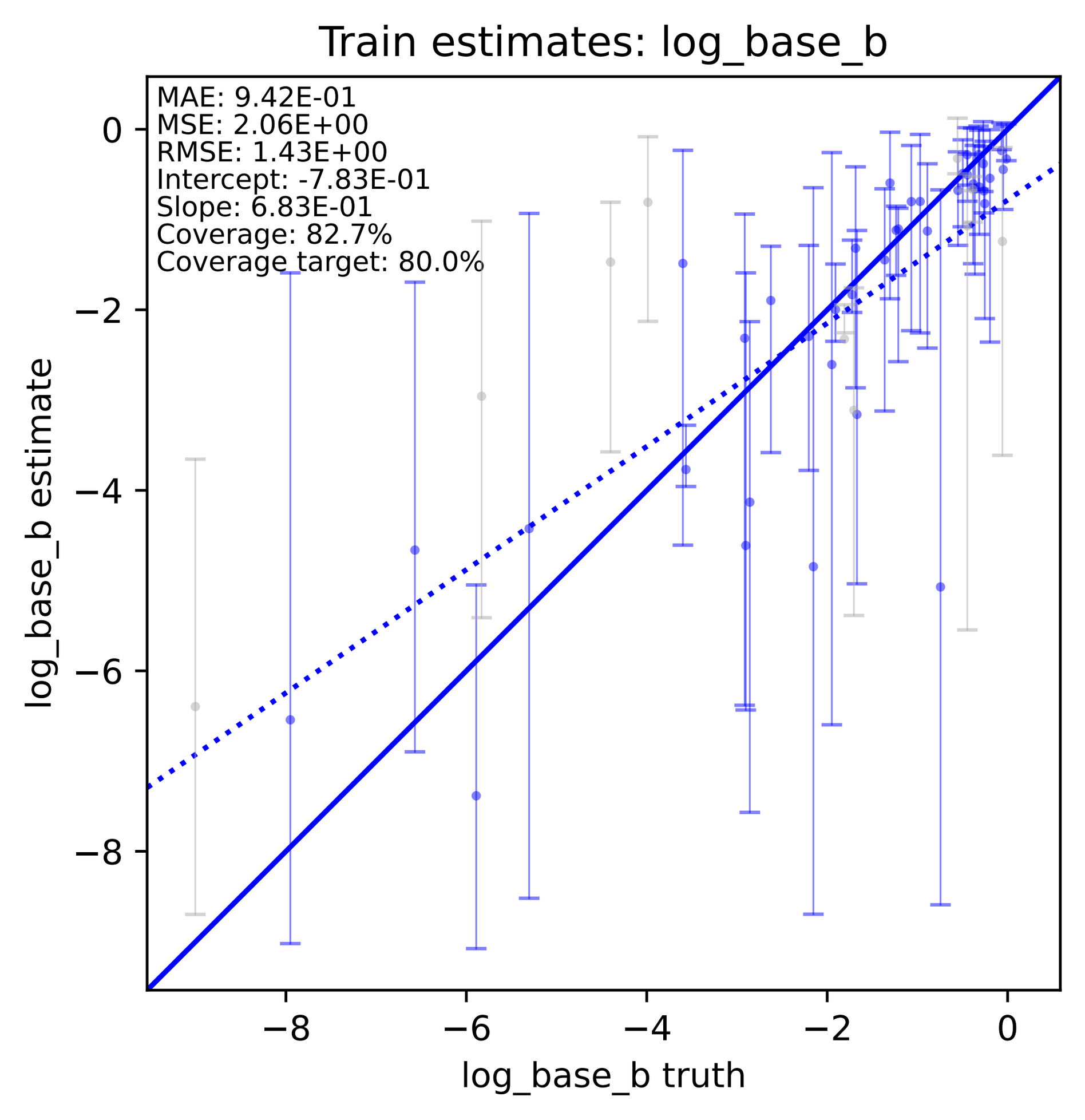

PCA: training phy. data
PC2 (9%)
% variance
PC
PC3 (4%)
PC4 (2%)
PC1 (27%)
PC2 (9%)
PC3 (4%)

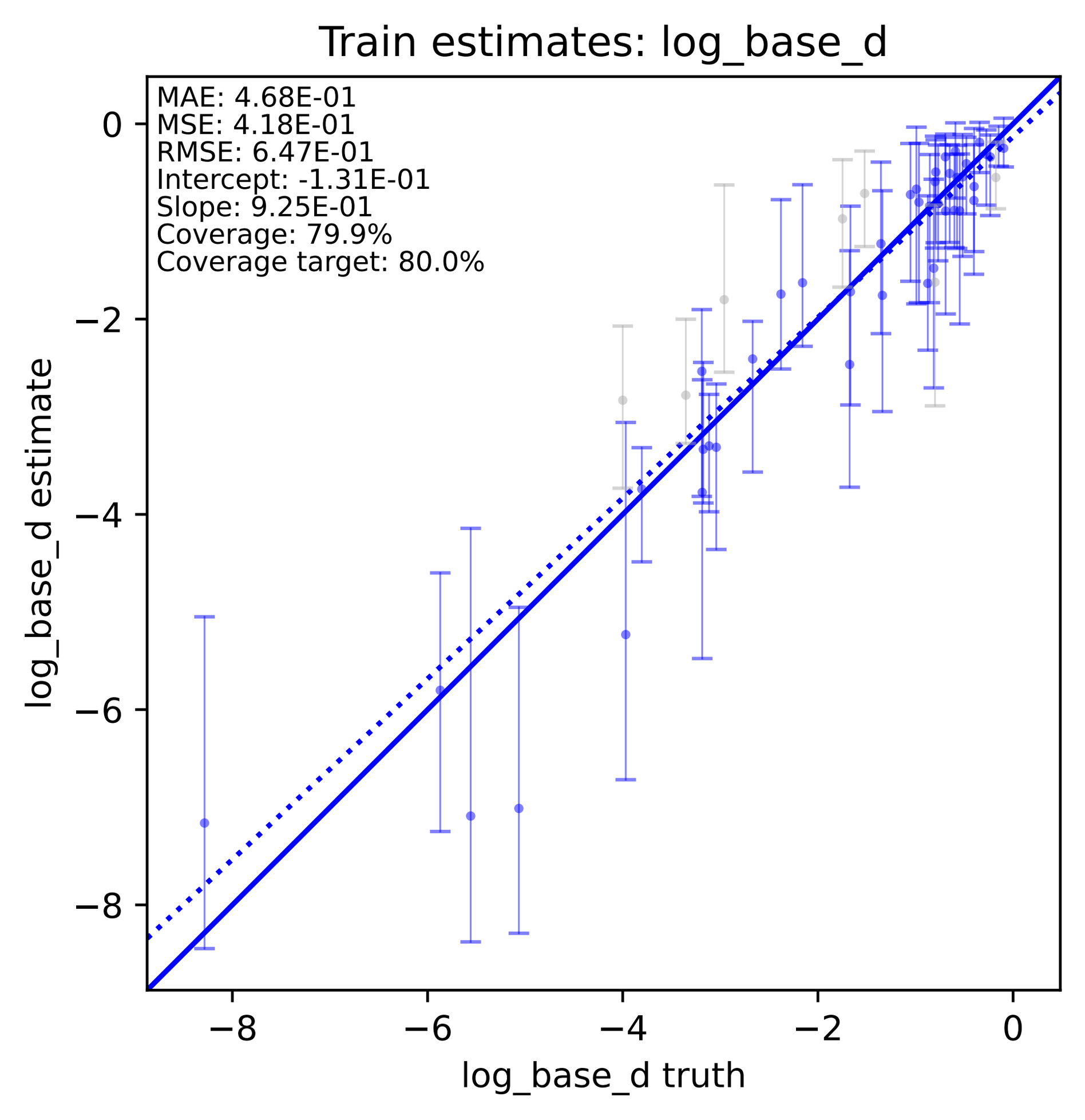

Density: training aux. data
N_bar
sim: 0.94 (0.59, 1.04)
emp: 0.99 (0.99, 0.99)
age_var
sim: 1.19 (-0.43, 2.76)
emp: 2.43 (2.43, 2.43)
brlen_var
sim: 0.72 (-0.99, 2.42)
emp: 1.97 (1.97, 1.97)
colless
sim: -1.25 (-1.60, -0.40)
emp: -1.22 (-1.22, -1.22)
f_dat_0
sim: 5.05 (1.57, 473.57)
emp: 4.00 (4.00, 4.00)
f_dat_1
sim: 2.24 (1.01, 118.88)
emp: 1.00 (1.00, 1.00)
log10_B1
sim: 1.84 (0.75, 2.19)
emp: 1.90 (1.90, 1.90)
log10_age_mean
sim: 0.50 (-0.19, 1.48)
emp: 1.36 (1.36, 1.36)
log10_brlen_mean
sim: 0.27 (-0.47, 1.23)
emp: 1.07 (1.07, 1.07)
log10_root_age
sim: 1.24 (0.46, 1.96)
emp: 1.81 (1.81, 1.81)
log10_tree_length
sim: 2.55 (1.47, 3.47)
emp: 3.56 (3.56, 3.56)
n_dat_0
sim: 6.89 (3.95, 10.04)
emp: 6.45 (6.45, 6.45)
n_dat_1
sim: 5.89 (3.33, 8.65)
emp: 5.06 (5.06, 5.06)
num_taxa
sim: 130.00 (11.00, 284.00)
emp: 158.00 (158.00, 158.00)
prop_taxa
sim: 1.00 (1.00, 1.00)
emp: 1.00 (1.00, 1.00)
treeness
sim: -0.28 (-0.40, -0.07)
emp: -0.44 (-0.44, -0.44)
Density
Variable

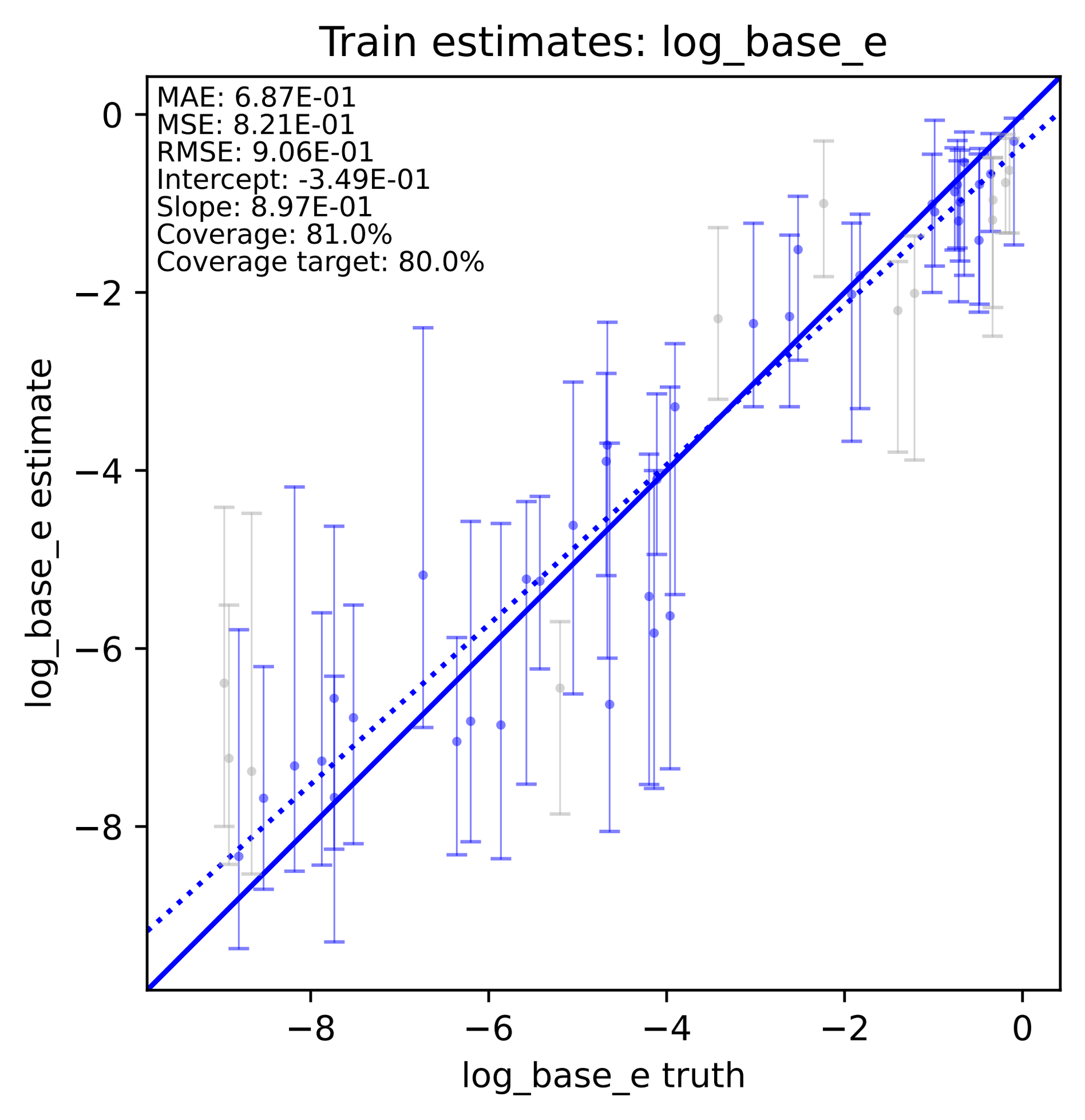

Density: training labels
exp_scalar_d_j
sim: 0.39 (0.14, 0.96)
emp: 0.21 (0.21, 0.21)
exp_scalar_e
sim: 2.33 (1.04, 6.89)
emp: 2.49 (2.49, 2.49)
log_base_b
sim: -1.52 (-8.65, -0.05)
emp: -2.25 (-2.25, -2.25)
log_base_d
sim: -1.17 (-8.44, -0.03)
emp: -6.71 (-6.71, -6.71)
log_base_e
sim: -2.93 (-8.87, -0.18)
emp: -8.27 (-8.27, -8.27)
log_base_w
sim: -1.08 (-7.90, -0.03)
emp: -3.16 (-3.16, -3.16)
Density
Variable

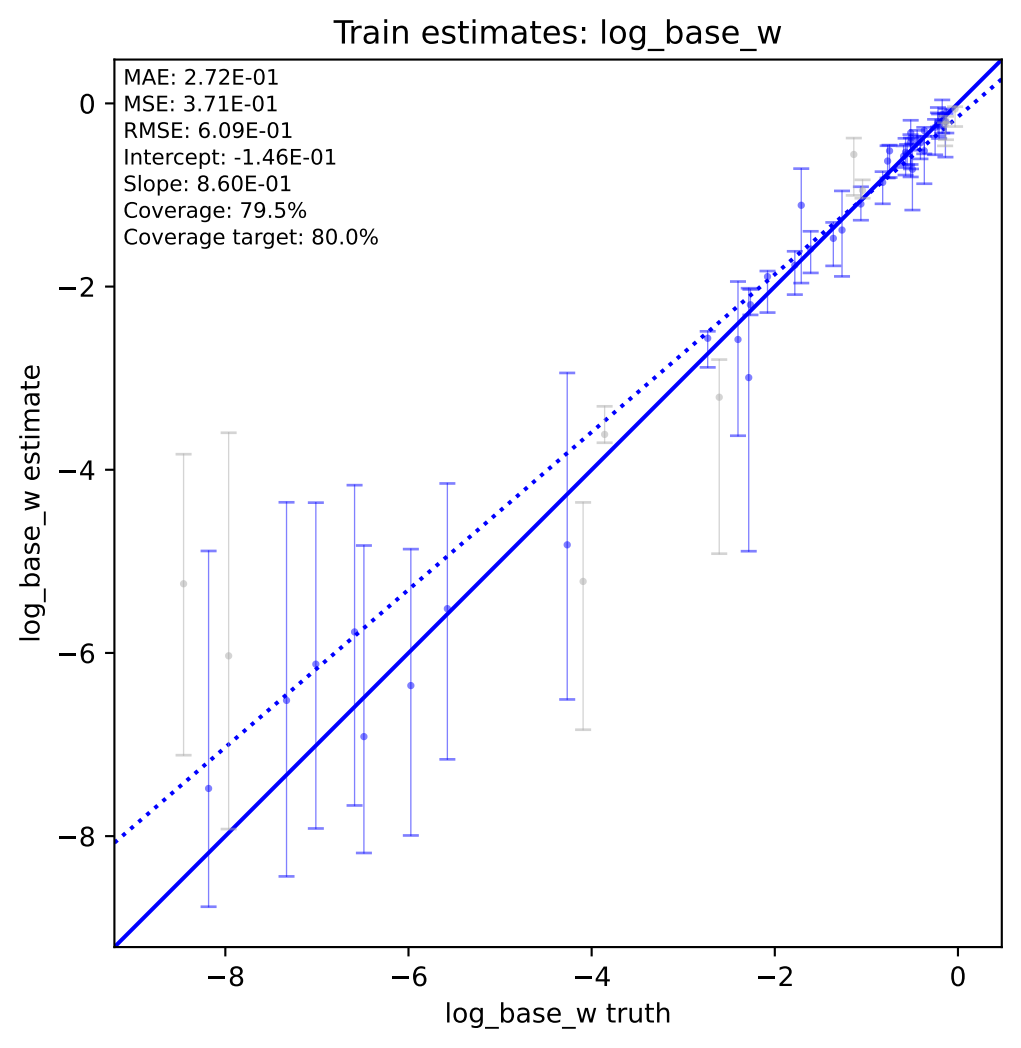

Train estimates: exp_scalar_d_j
MAE: 1.42E-01
MSE: 3.62E-02
RMSE: 1.90E-01
Intercept: 2.20E-01
Slope: 4.26E-01
Coverage: 80.4%
Coverage target: 80.0%
exp_scalar_d_j estimate
exp_scalar_d_j truth
0.2
0.4
0.6
0.8
1.0

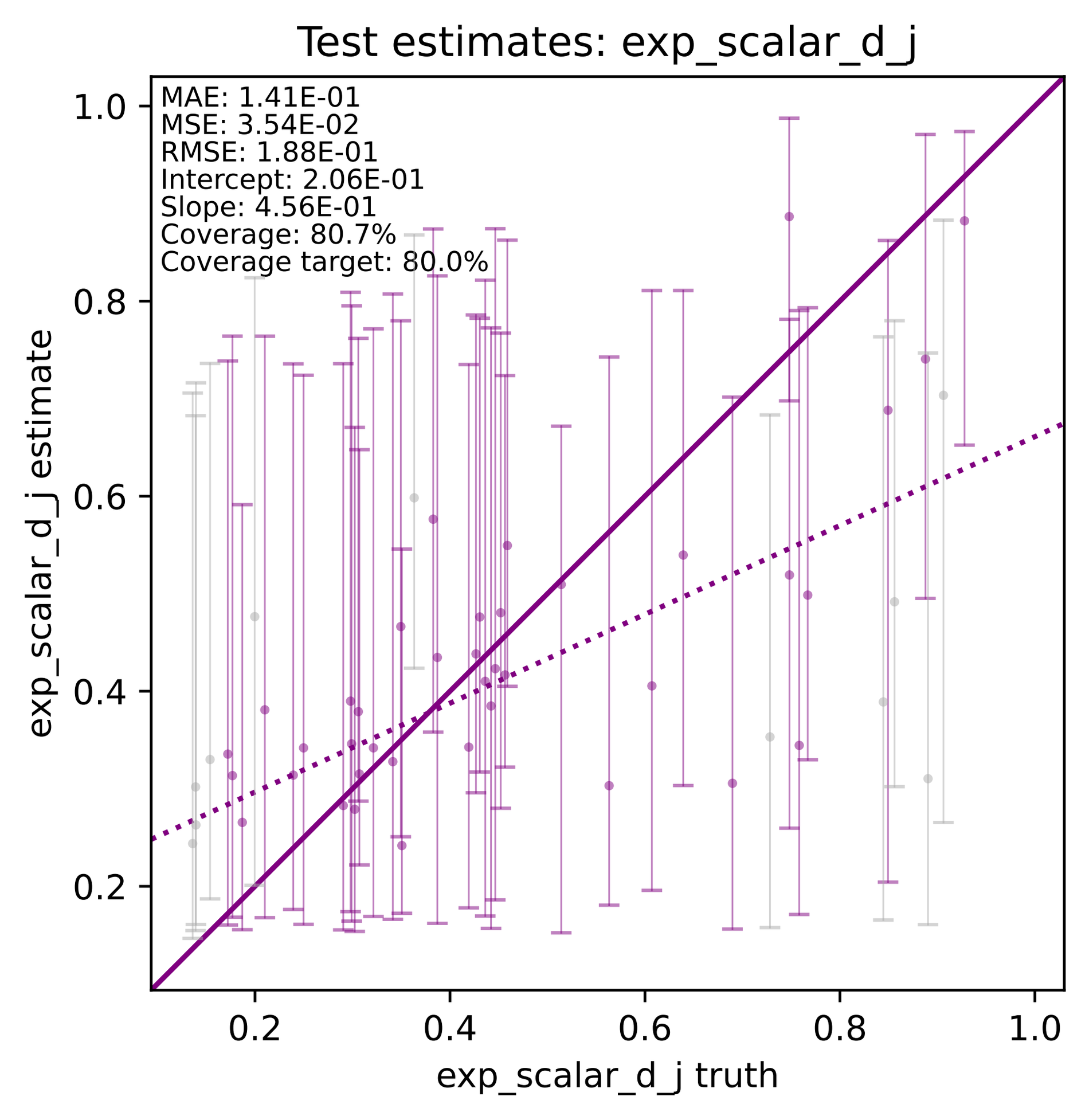

Train estimates: exp_scalar_e
MAE: 1.21E+00
MSE: 2.66E+00
RMSE: 1.63E+00
Intercept: 2.03E+00
Slope: 1.42E-01
Coverage: 79.8%
Coverage target: 80.0%
exp_scalar_e estimate
exp_scalar_e truth
1
2
3
4
5
6
7

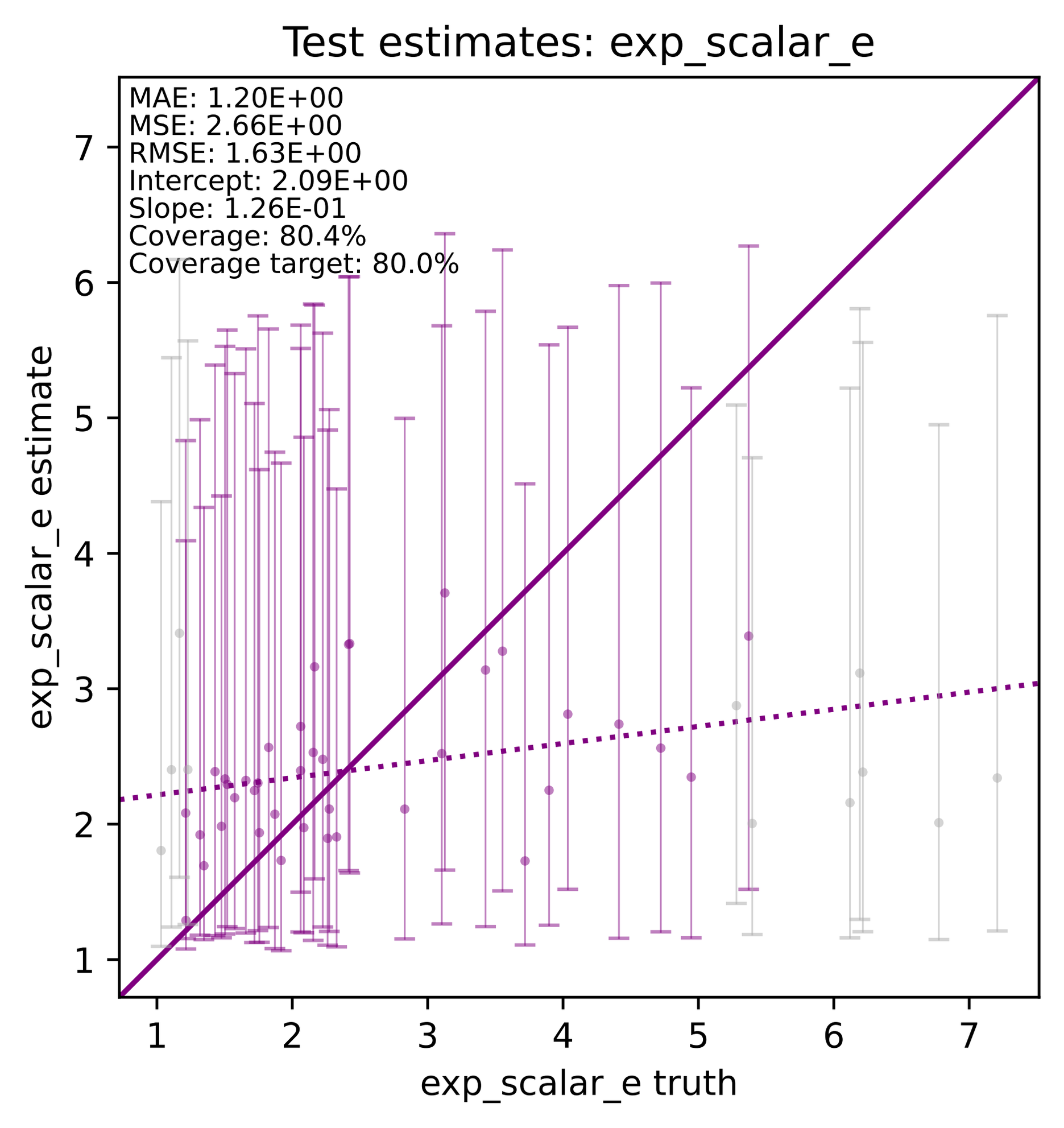

Train estimates: log_base_b
MAE: 9.42E-01
MSE: 2.06E+00
RMSE: 1.43E+00
Intercept: -7.83E-01
Slope: 6.83E-01
Coverage: 82.7%
Coverage target: 80.0%
log_base_b estimate
log_base_b truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

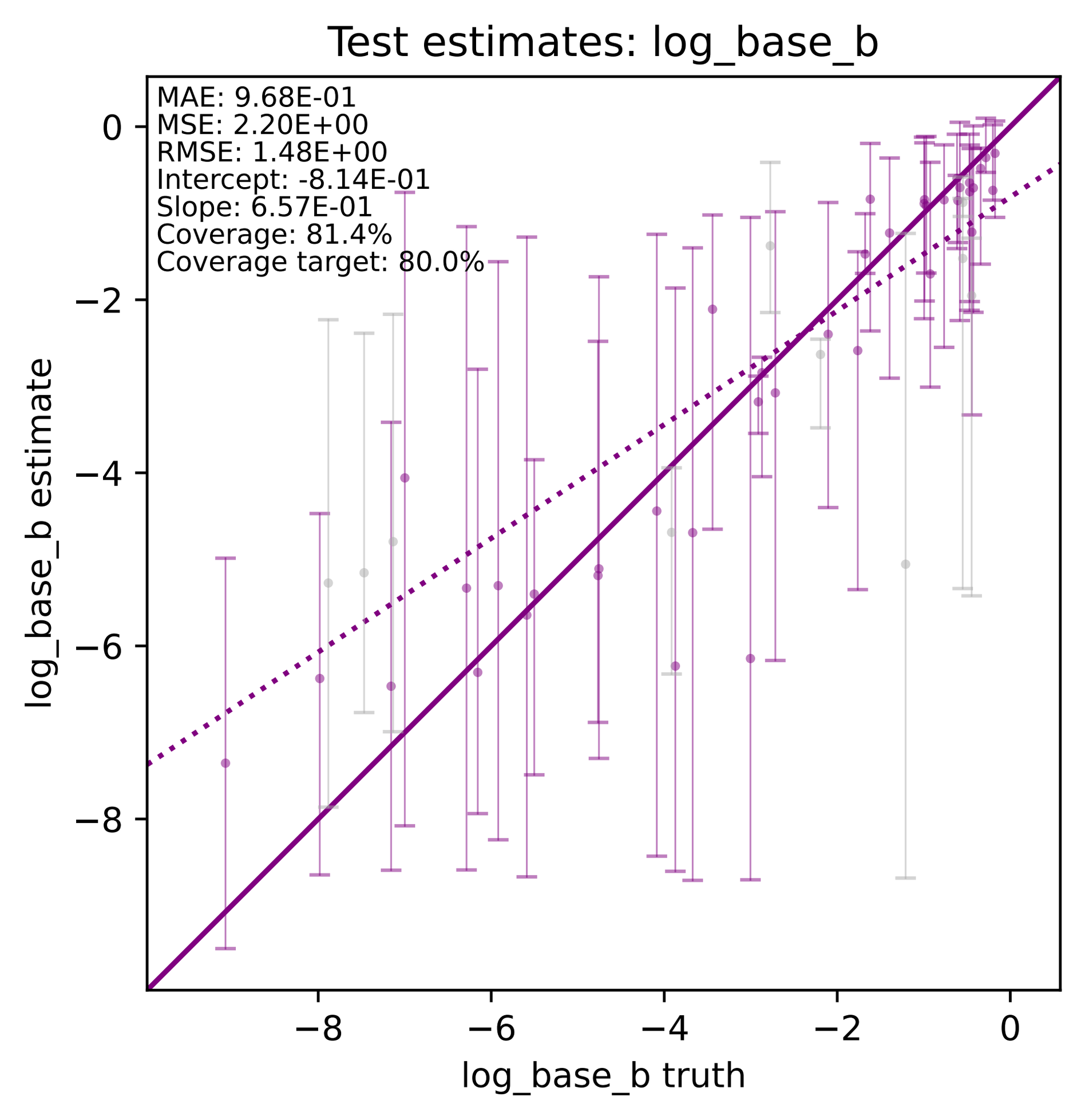

Train estimates: log_base_d
MAE: 4.68E-01
MSE: 4.18E-01
RMSE: 6.47E-01
Intercept: -1.31E-01
Slope: 9.25E-01
Coverage: 79.9%
Coverage target: 80.0%
log_base_d estimate
log_base_d truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

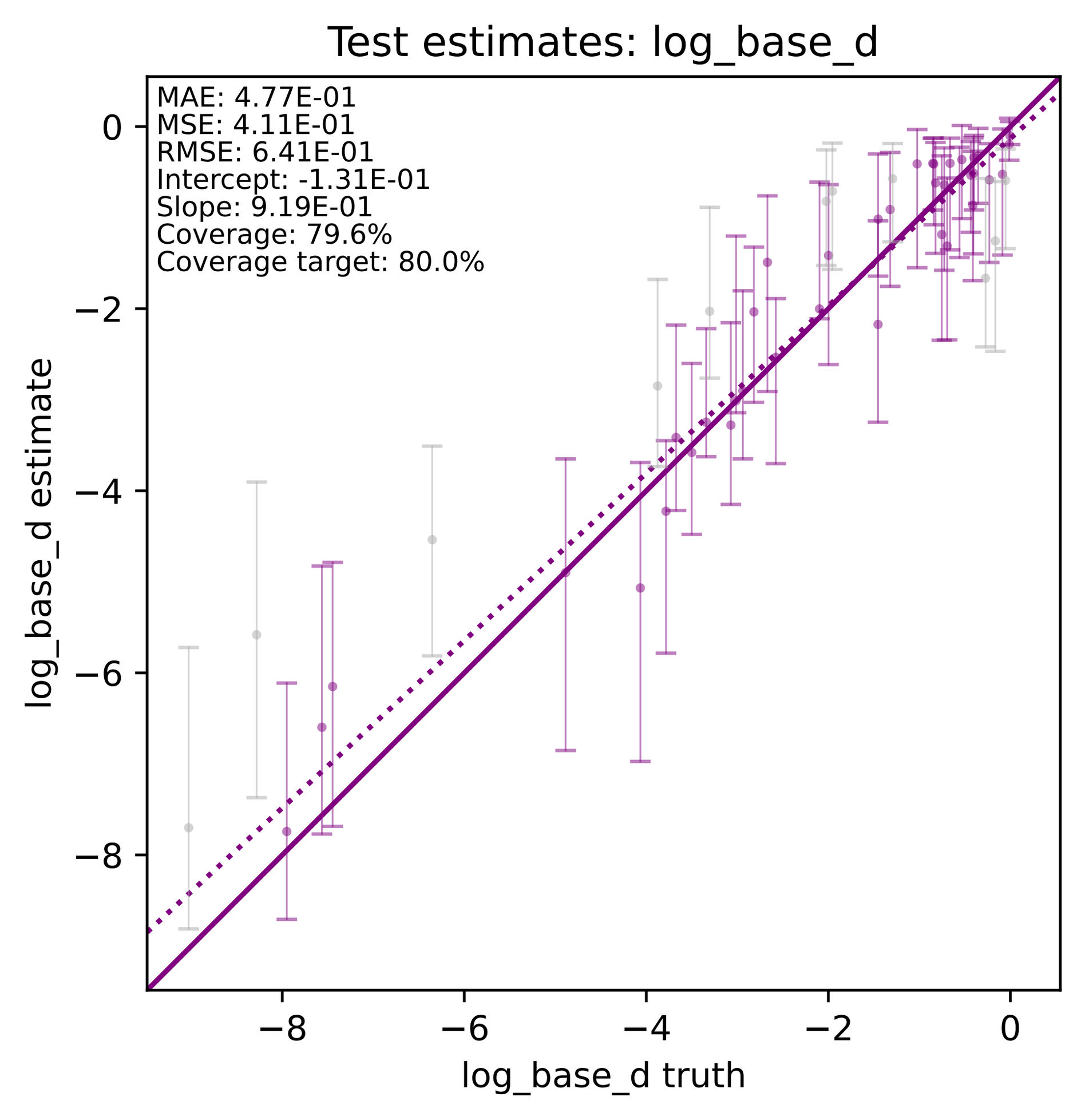

Train estimates: log_base_e
MAE: 6.87E-01
MSE: 8.21E-01
RMSE: 9.06E-01
Intercept: -3.49E-01
Slope: 8.97E-01
Coverage: 81.0%
Coverage target: 80.0%
log_base_e estimate
log_base_e truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

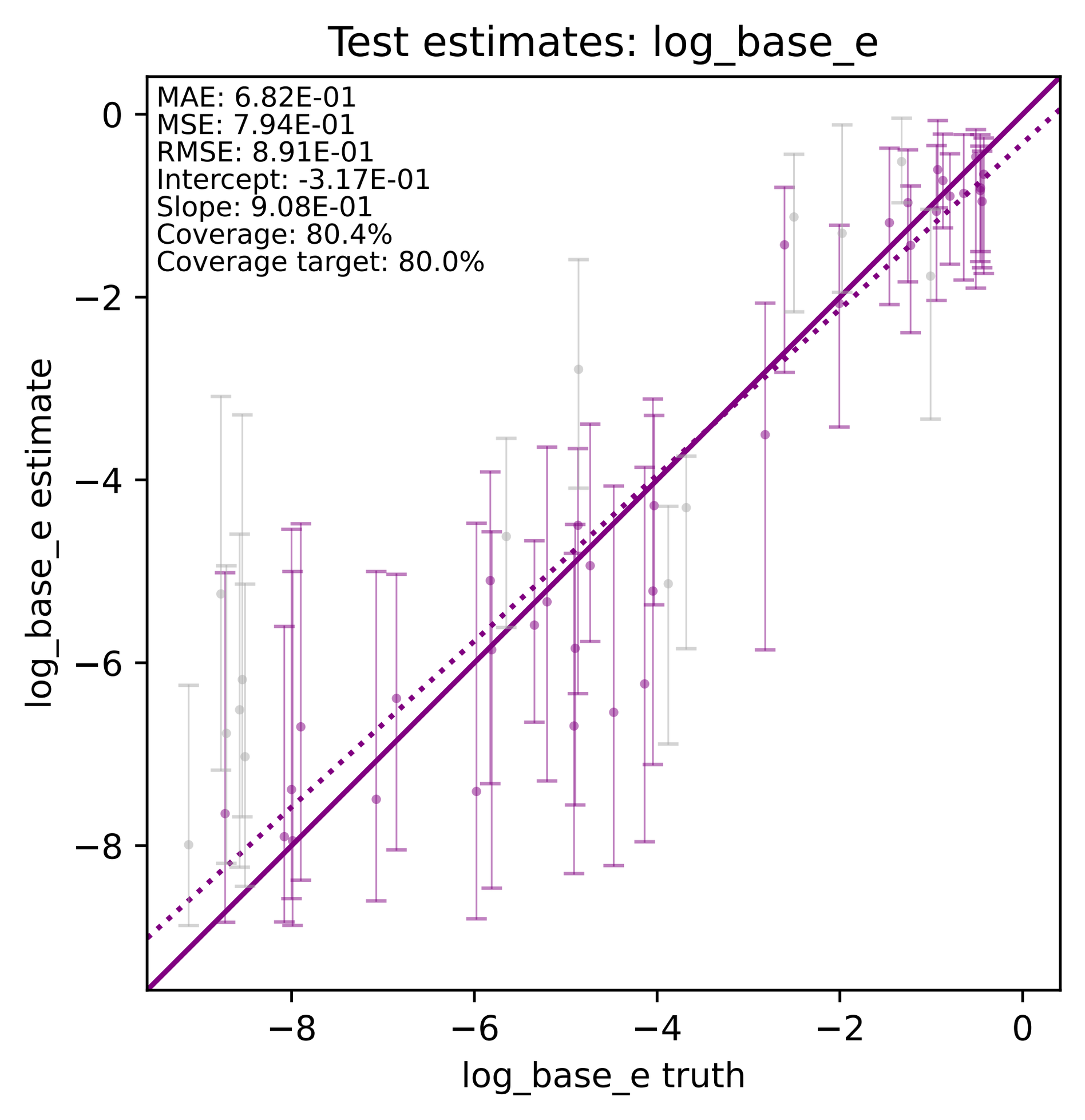

Train estimates: log_base_w
MAE: 2.72E-01
MSE: 3.71E-01
RMSE: 6.09E-01
Intercept: -1.46E-01
Slope: 8.60E-01
Coverage: 79.5%
Coverage target: 80.0%
log_base_w estimate
log_base_w truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

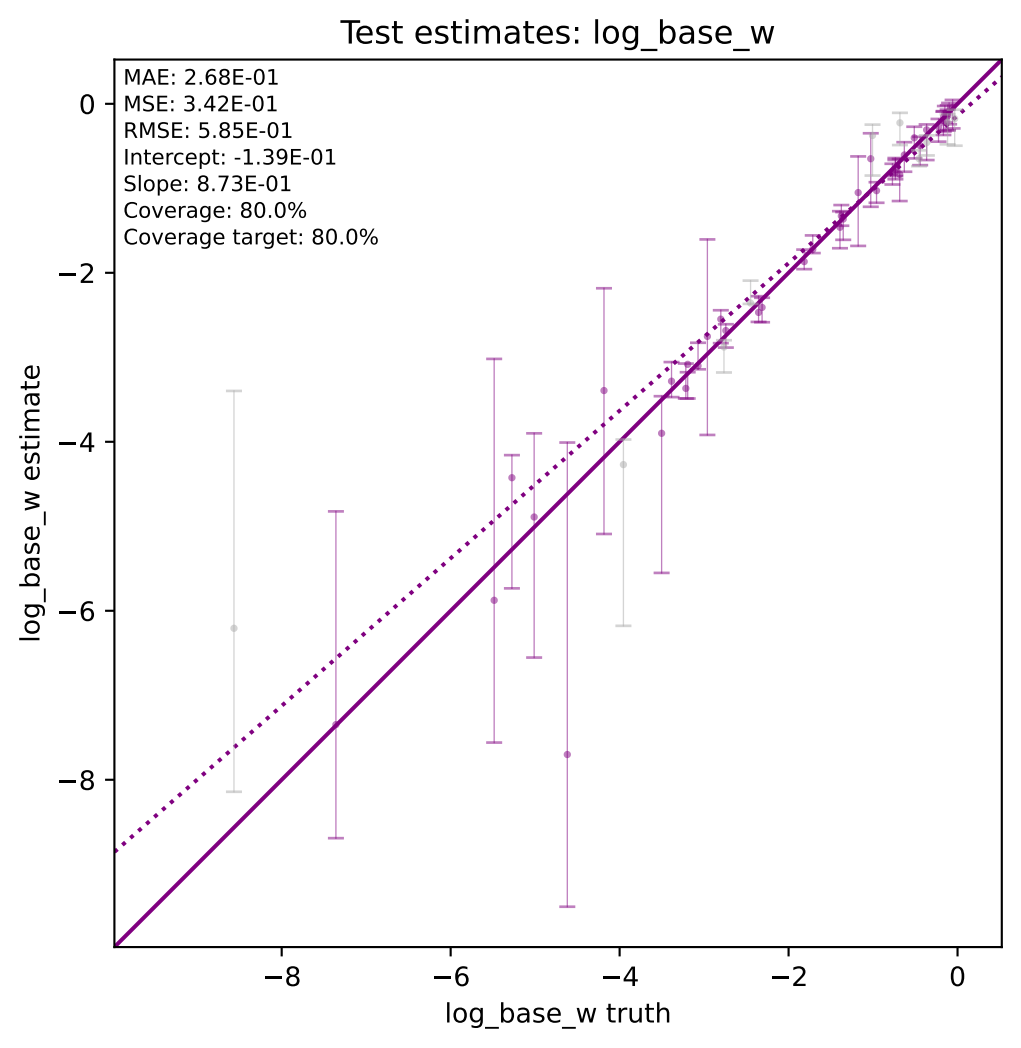

Test estimates: exp_scalar_d_j
MAE: 1.41E-01
MSE: 3.54E-02
RMSE: 1.88E-01
Intercept: 2.06E-01
Slope: 4.56E-01
Coverage: 80.7%
Coverage target: 80.0%
exp_scalar_d_j estimate
exp_scalar_d_j truth
0.2
0.4
0.6
0.8
1.0

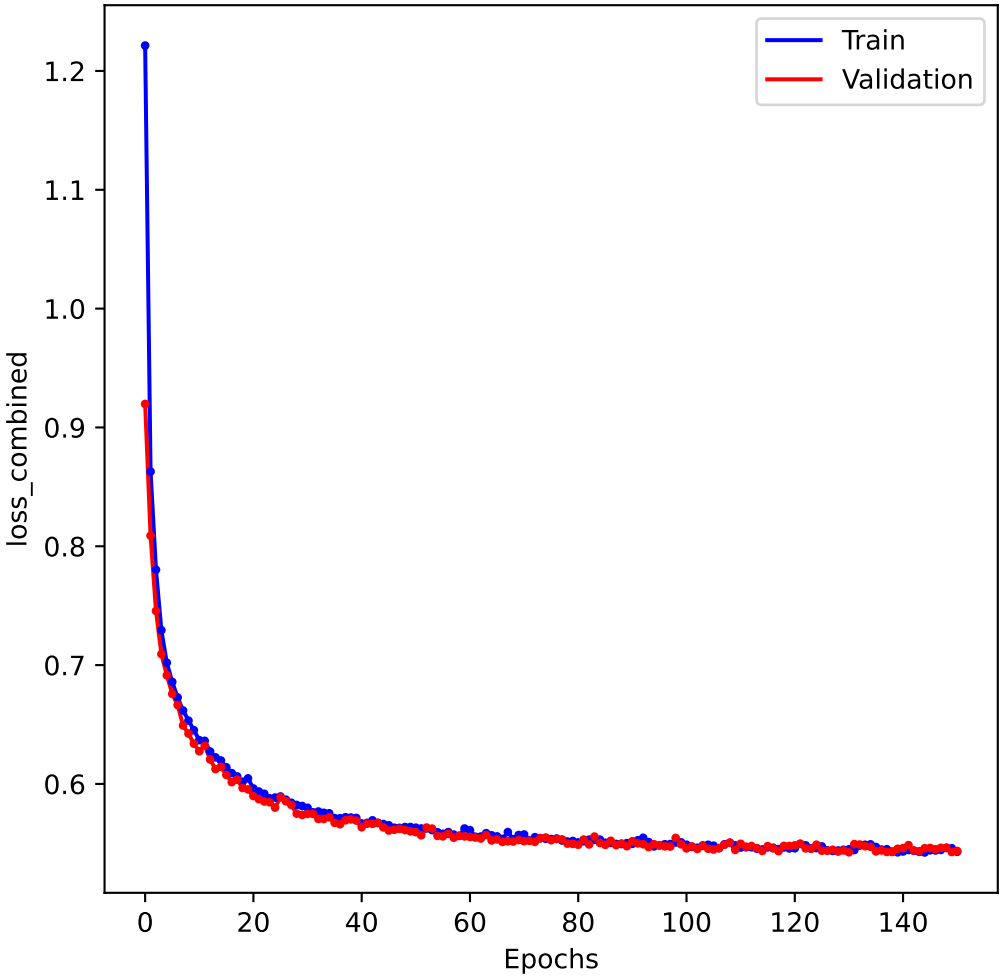

Test estimates: exp_scalar_e
MAE: 1.20E+00
MSE: 2.66E+00
RMSE: 1.63E+00
Intercept: 2.09E+00
Slope: 1.26E-01
Coverage: 80.4%
Coverage target: 80.0%
exp_scalar_e estimate
exp_scalar_e truth
1
2
3
4
5
6
7

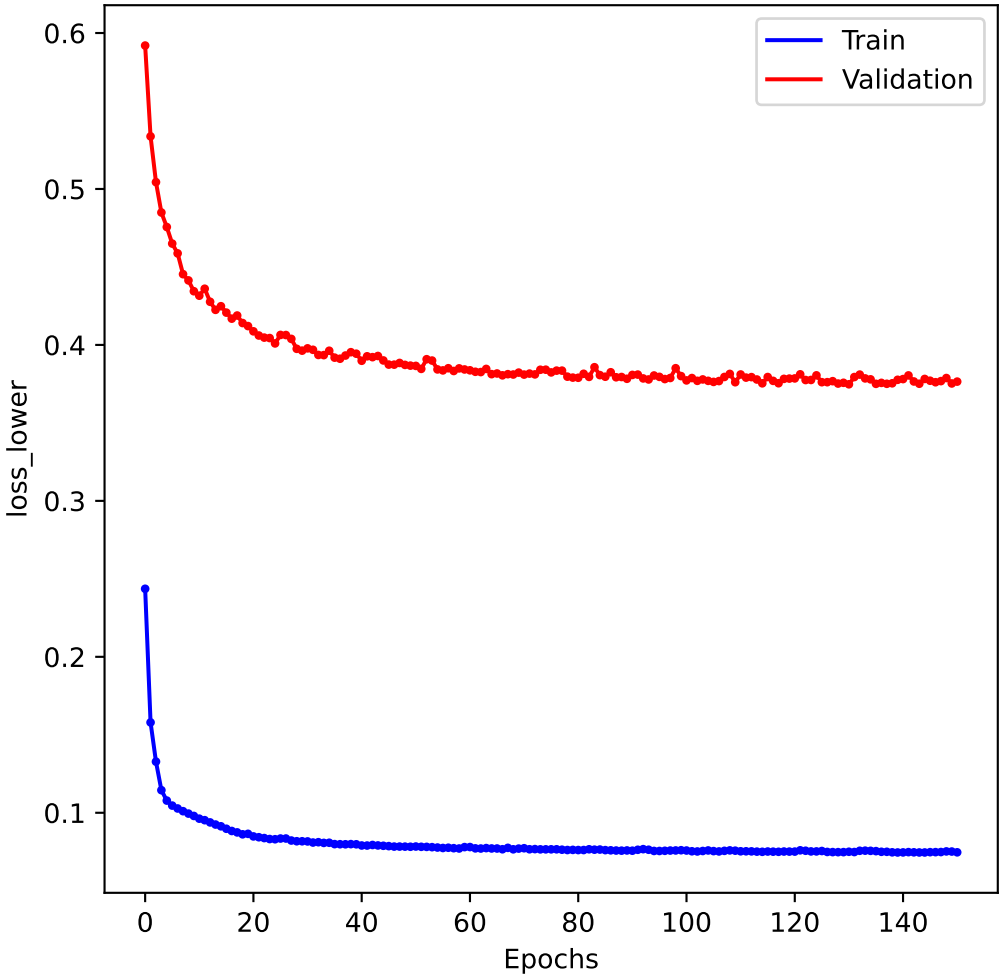

Test estimates: log_base_b
MAE: 9.68E-01
MSE: 2.20E+00
RMSE: 1.48E+00
Intercept: -8.14E-01
Slope: 6.57E-01
Coverage: 81.4%
Coverage target: 80.0%
log_base_b estimate
log_base_b truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

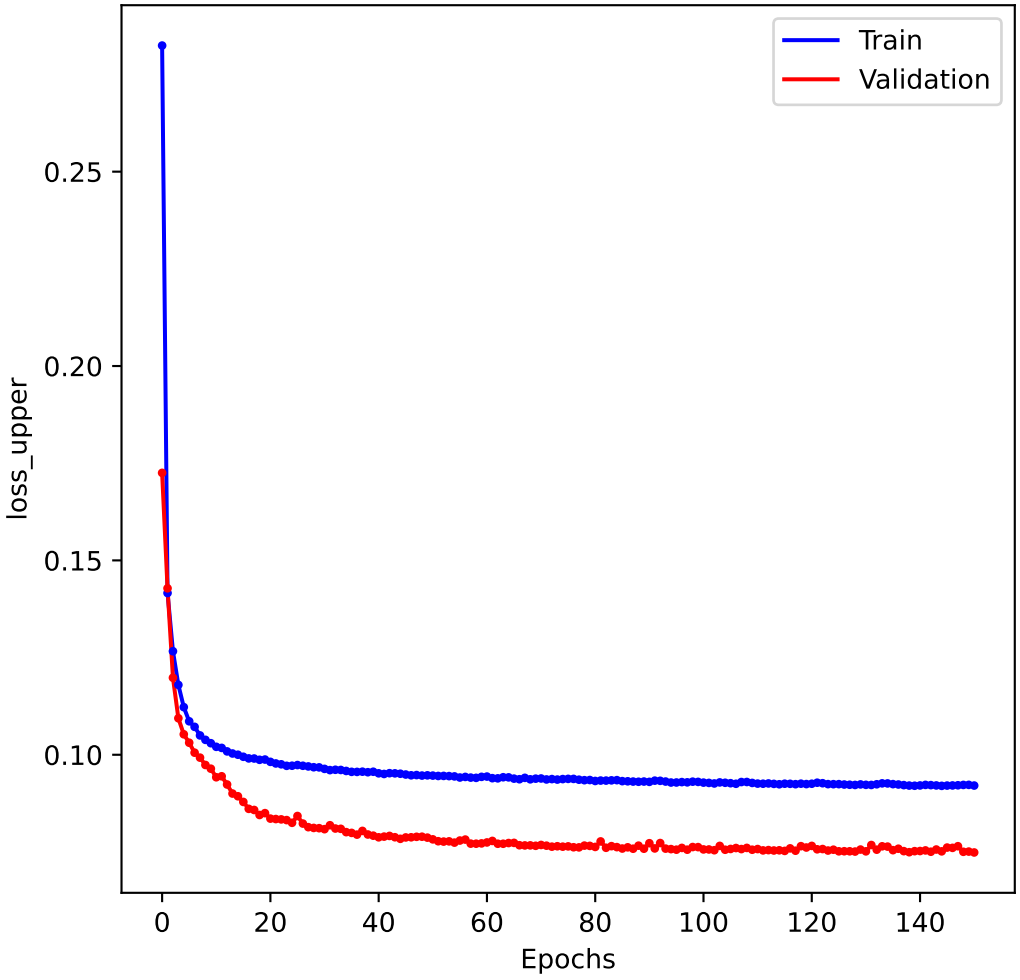

Test estimates: log_base_d
MAE: 4.77E-01
MSE: 4.11E-01
RMSE: 6.41E-01
Intercept: -1.31E-01
Slope: 9.19E-01
Coverage: 79.6%
Coverage target: 80.0%
log_base_d estimate
log_base_d truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

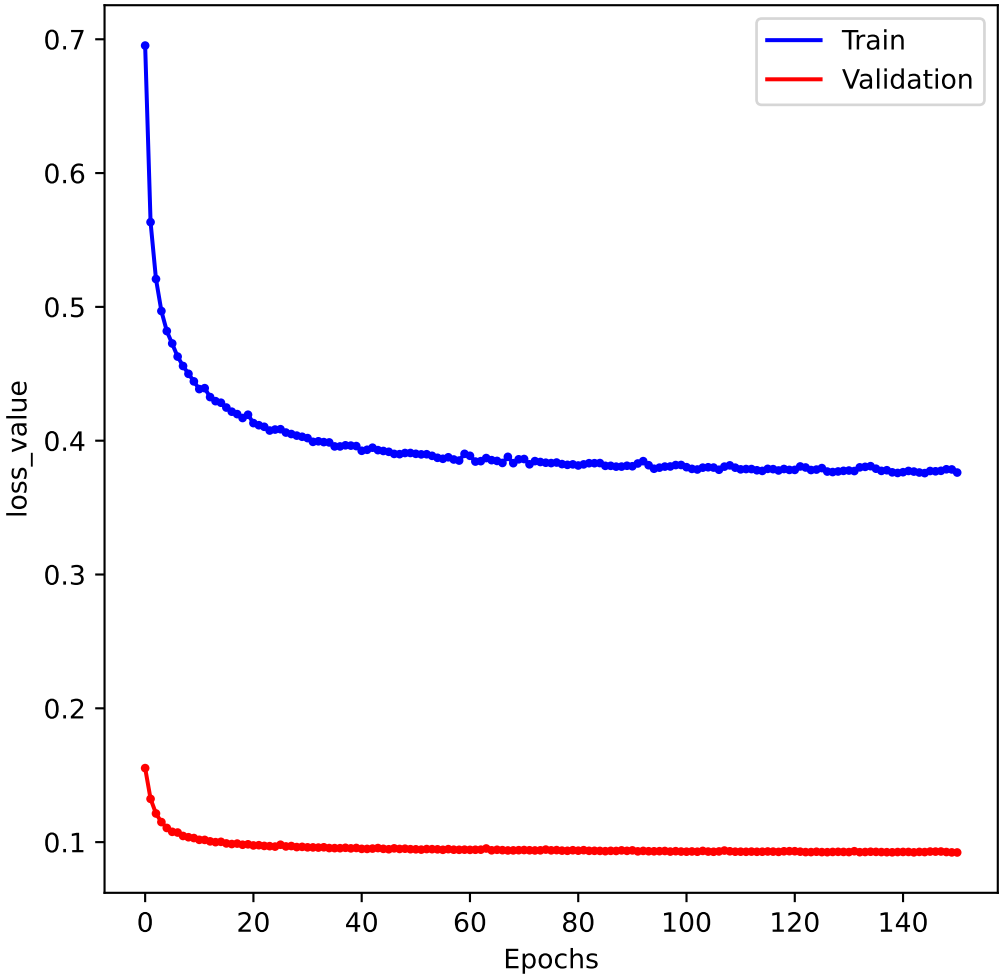

Test estimates: log_base_e
MAE: 6.82E-01
MSE: 7.94E-01
RMSE: 8.91E-01
Intercept: -3.17E-01
Slope: 9.08E-01
Coverage: 80.4%
Coverage target: 80.0%
log_base_e estimate
log_base_e truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

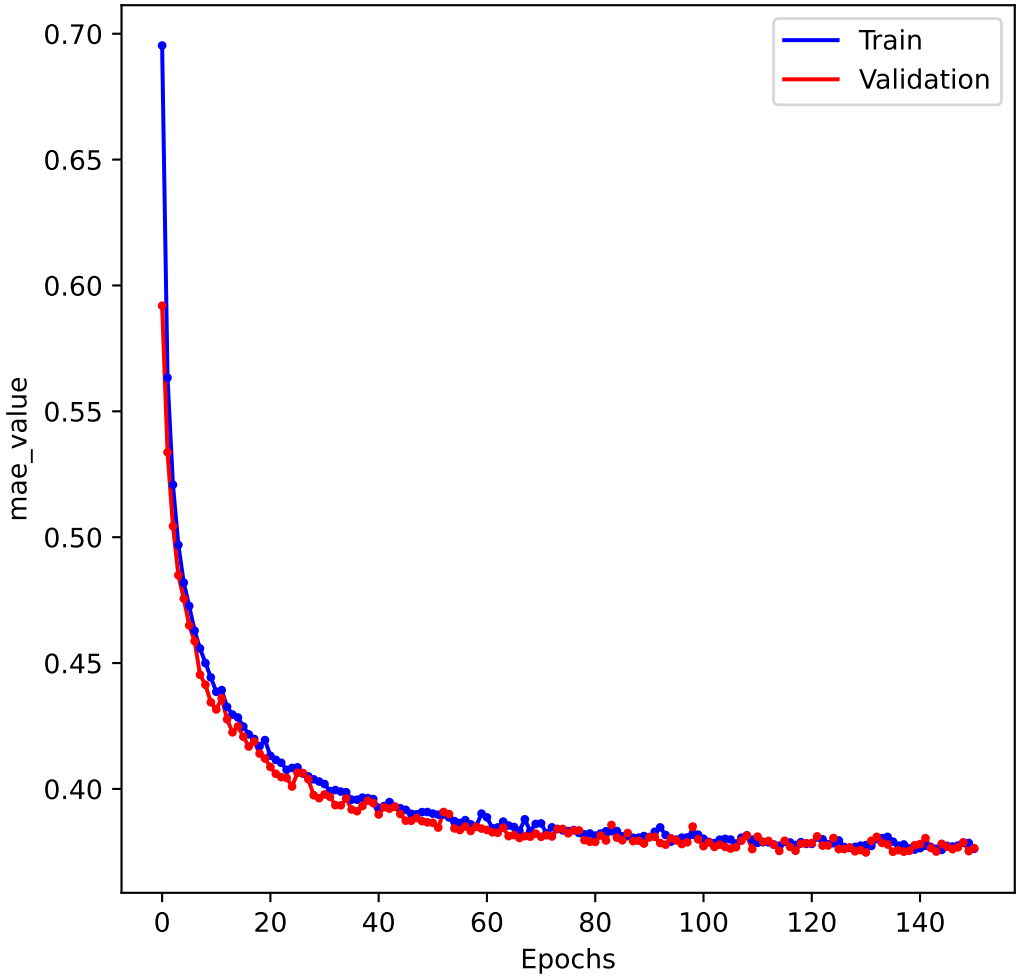

Test estimates: log_base_w
MAE: 2.68E-01
MSE: 3.42E-01
RMSE: 5.85E-01
Intercept: -1.39E-01
Slope: 8.73E-01
Coverage: 80.0%
Coverage target: 80.0%
log_base_w estimate
log_base_w truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

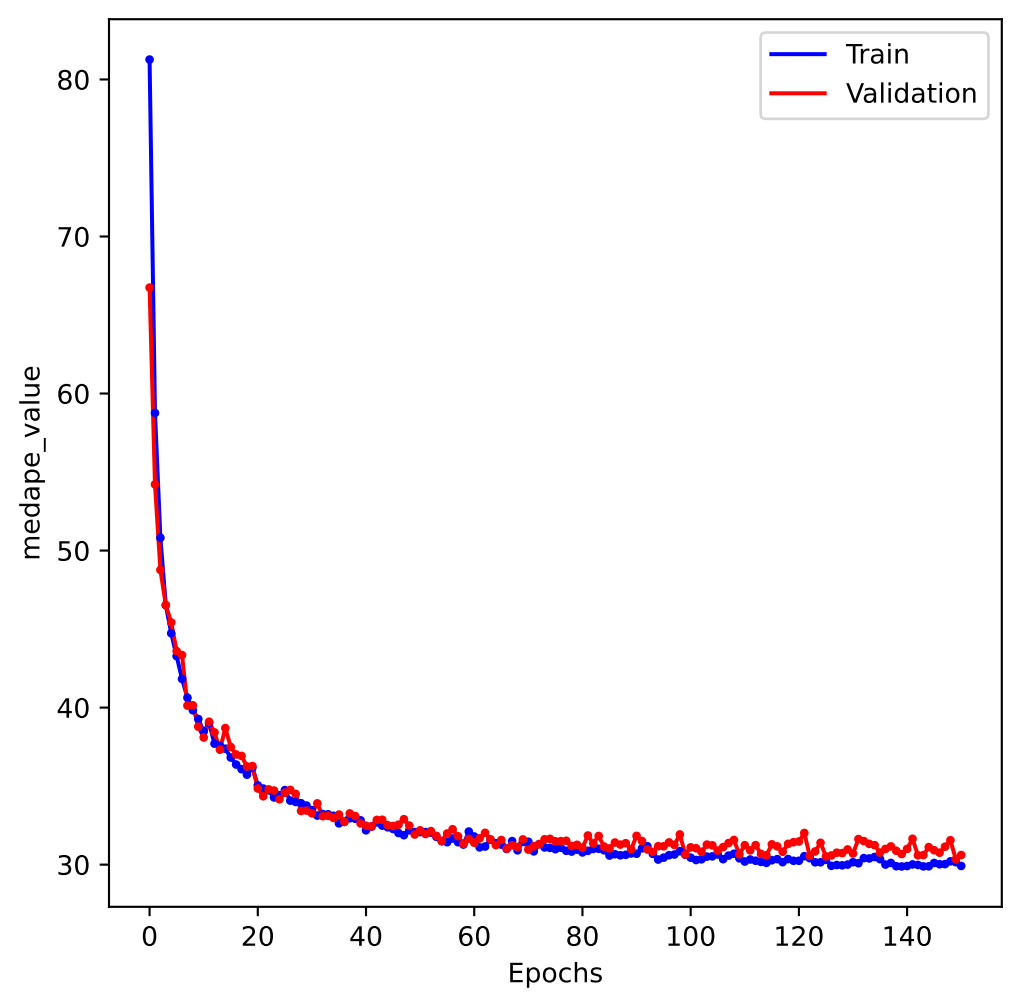

Train
Validation
loss_combined
1.2
1.1
1.0
0.9
0.8
0.7
0.6
0
20
40
60
80
100
120
140
Epochs

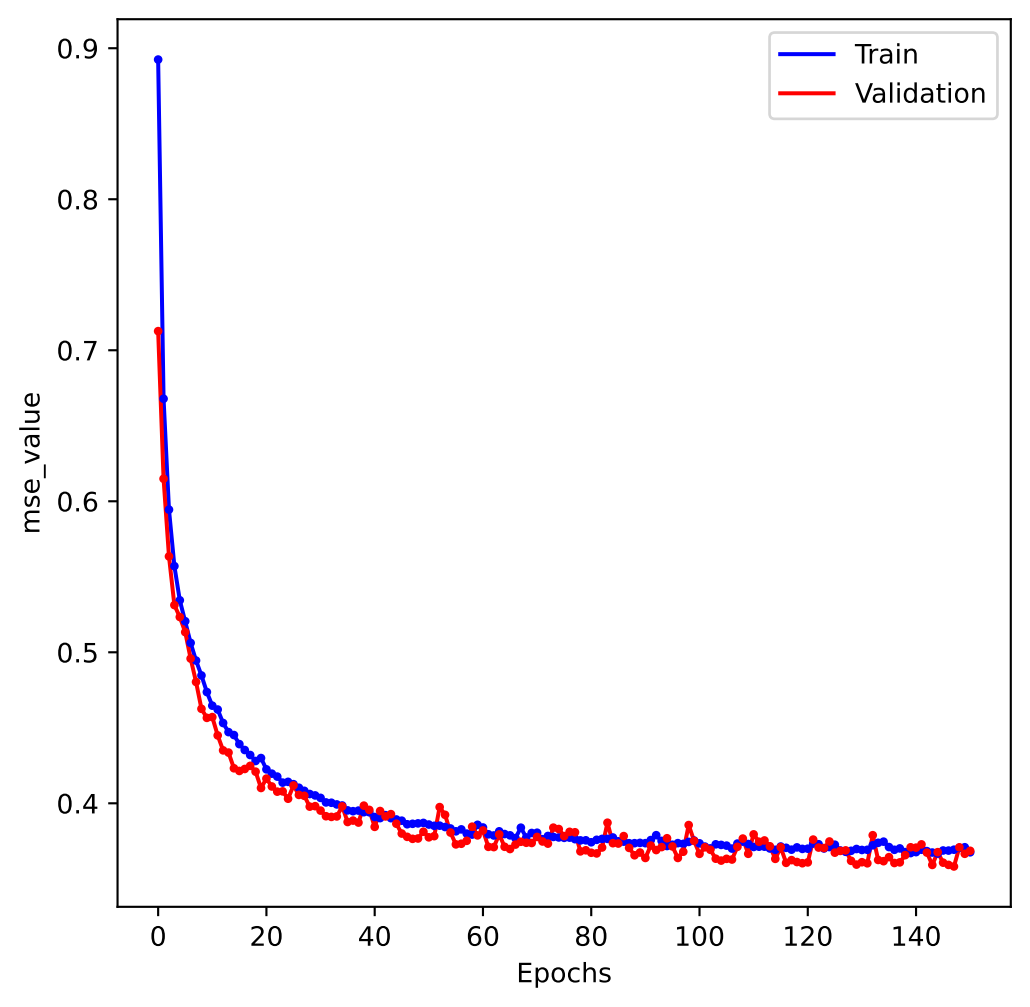

Train
Validation
loss_lower
0.6
0.5
0.4
0.3
0.2
0.1
0
20
40
60
80
100
120
140
Epochs

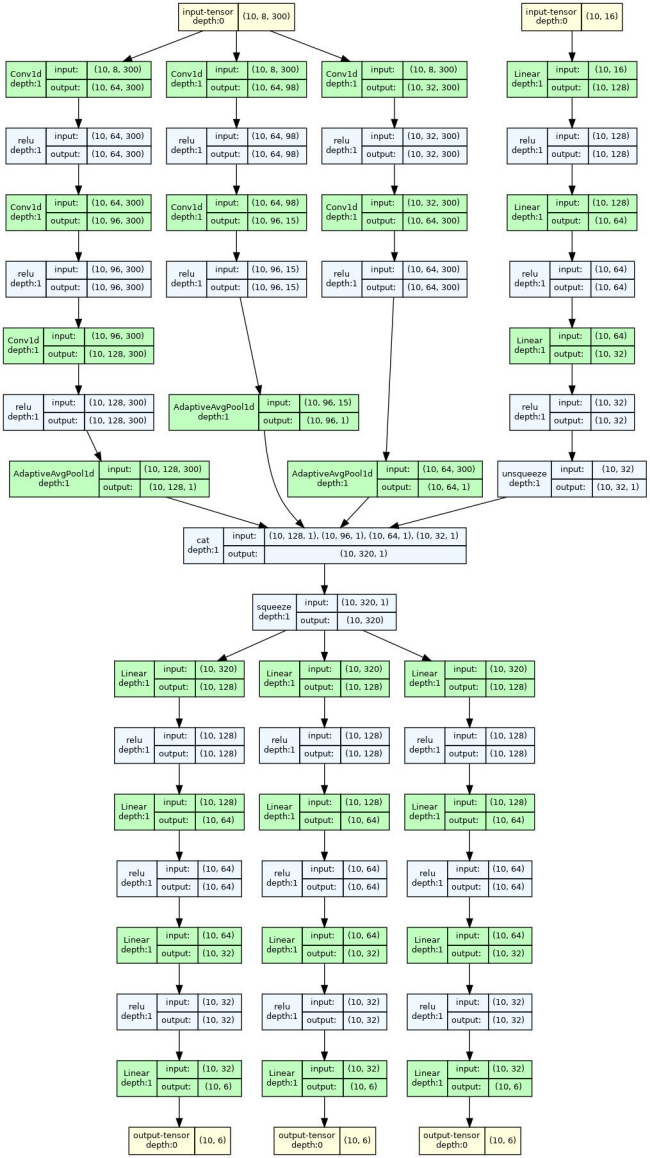

Train
Validation
0.25
0.20
0.15
0.10
loss_upper
0
20
40
60
80
100
120
140
Epochs


Train
Validation
loss_value
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
20
40
60
80
100
120
140
Epochs


Train
Validation
mae_value
0.70
0.65
0.60
0.55
0.50
0.45
0.40
0
20
40
60
80
100
120
140
Epochs


Train
Validation
medape_value
Epochs
80
70
60
50
40
30
0
20
40
60
80
100
120
140


Train
Validation
mse_value
0.9
0.8
0.7
0.6
0.5
0.4
0
20
40
60
80
100
120
140
Epochs


input-tensor depth:0
(10, 8, 300)
input-tensor depth:0
(10, 16)
Conv1d depth:1
input: (10, 8, 300)
output: (10, 64, 300)
Conv1d depth:1
input: (10, 8, 300)
output: (10, 64, 98)
Conv1d depth:1
input: (10, 8, 300)
output: (10, 32, 300)
Linear depth:1
input: (10, 16)
output: (10, 128)
relu depth:1
input: (10, 64, 300)
output: (10, 64, 300)
relu depth:1
input: (10, 64, 98)
output: (10, 64, 98)
relu depth:1
input: (10, 32, 300)
output: (10, 32, 300)
relu depth:1
input: (10, 128)
output: (10, 128)
Conv1d depth:1
input: (10, 64, 300)
output: (10, 96, 300)
Conv1d depth:1
input: (10, 64, 98)
output: (10, 96, 15)
Conv1d depth:1
input: (10, 32, 300)
output: (10, 64, 300)
Linear depth:1
input: (10, 128)
output: (10, 64)
relu depth:1
input: (10, 96, 300)
output: (10, 96, 300)
relu depth:1
input: (10, 96, 15)
output: (10, 96, 15)
relu depth:1
input: (10, 64, 300)
output: (10, 64, 300)
relu depth:1
input: (10, 64)
output: (10, 64)
Conv1d depth:1
input: (10, 96, 300)
output: (10, 128, 300)
Linear depth:1
input: (10, 64)
output: (10, 32)
relu depth:1
input: (10, 128, 300)
output: (10, 128, 300)
AdaptiveAvgPool1d depth:1
input: (10, 96, 15)
output: (10, 96, 1)
relu depth:1
input: (10, 32)
output: (10, 32)
AdaptiveAvgPool1d depth:1
input: (10, 128, 300)
output: (10, 128, 1)
AdaptiveAvgPool1d depth:1
input: (10, 64, 300)
output: (10, 64, 1)
unsqueeze depth:1
input: (10, 32)
output: (10, 32, 1)
cat depth:1
input: (10, 128, 1), (10, 96, 1), (10, 64, 1), (10, 32, 1)
output: (10, 320, 1)
squeeze depth:1
input: (10, 320, 1)
output: (10, 320)
Linear depth:1
input: (10, 320)
output: (10, 128)
Linear depth:1
input: (10, 320)
output: (10, 128)
Linear depth:1
input: (10, 320)
output: (10, 128)
relu depth:1
input: (10, 128)
output: (10, 128)
relu depth:1
input: (10, 128)
output: (10, 128)
relu depth:1
input: (10, 128)
output: (10, 128)
Linear depth:1
input: (10, 128)
output: (10, 64)
Linear depth:1
input: (10, 128)
output: (10, 64)
Linear depth:1
input: (10, 128)
output: (10, 64)
relu depth:1
input: (10, 64)
output: (10, 64)
relu depth:1
input: (10, 64)
output: (10, 64)
relu depth:1
input: (10, 64)
output: (10, 64)
Linear depth:1
input: (10, 64)
output: (10, 32)
Linear depth:1
input: (10, 64)
output: (10, 32)
Linear depth:1
input: (10, 64)
output: (10, 32)
relu depth:1
input: (10, 32)
output: (10, 32)
relu depth:1
input: (10, 32)
output: (10, 32)
relu depth:1
input: (10, 32)
output: (10, 32)
Linear depth:1
input: (10, 32)
output: (10, 6)
Linear depth:1
input: (10, 32)
output: (10, 6)
Linear depth:1
input: (10, 32)
output: (10, 6)
output-tensor depth:0
(10, 6)
output-tensor depth:0
(10, 6)
output-tensor depth:0
(10, 6)